Getting started with Object Detection Using YOLO

Making Machines Detect Stuff Visually!

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Think About This...

"How does your phone camera instantly recognize faces?"

"How do self-driving cars avoid obstacles?"

"How does our AUV navigate underwater and identify objects?"

The answer: Object Detection - and today we'll learn how to build this superpower into our robots! 🚀

Prerequisites

What you need to know:

1. **Basic Python Programming**
2. **Basic CV Concepts**
3. **Curiosity to learn!**

No deep learning background needed - we'll keep it practical!

What is Object Detection?

1. What’s in the Image?

   Object detection identifies ***what*** objects are present (e.g., cat, car) *and* ***where*** they are (using bounding boxes).

1. Like a Smart Highlighter!

   It draws boxes around the objects and labels them!

1. Not Just Recognition!

   Unlike simple image classification (Which only says “This is a cat”), it also locates ***multiple*** objects in one image!

Object Detection = Finding + Identifying

🎯 Detection

"There's something here!"

🏷️ Classification

"It's a fish!"

📍 Localization

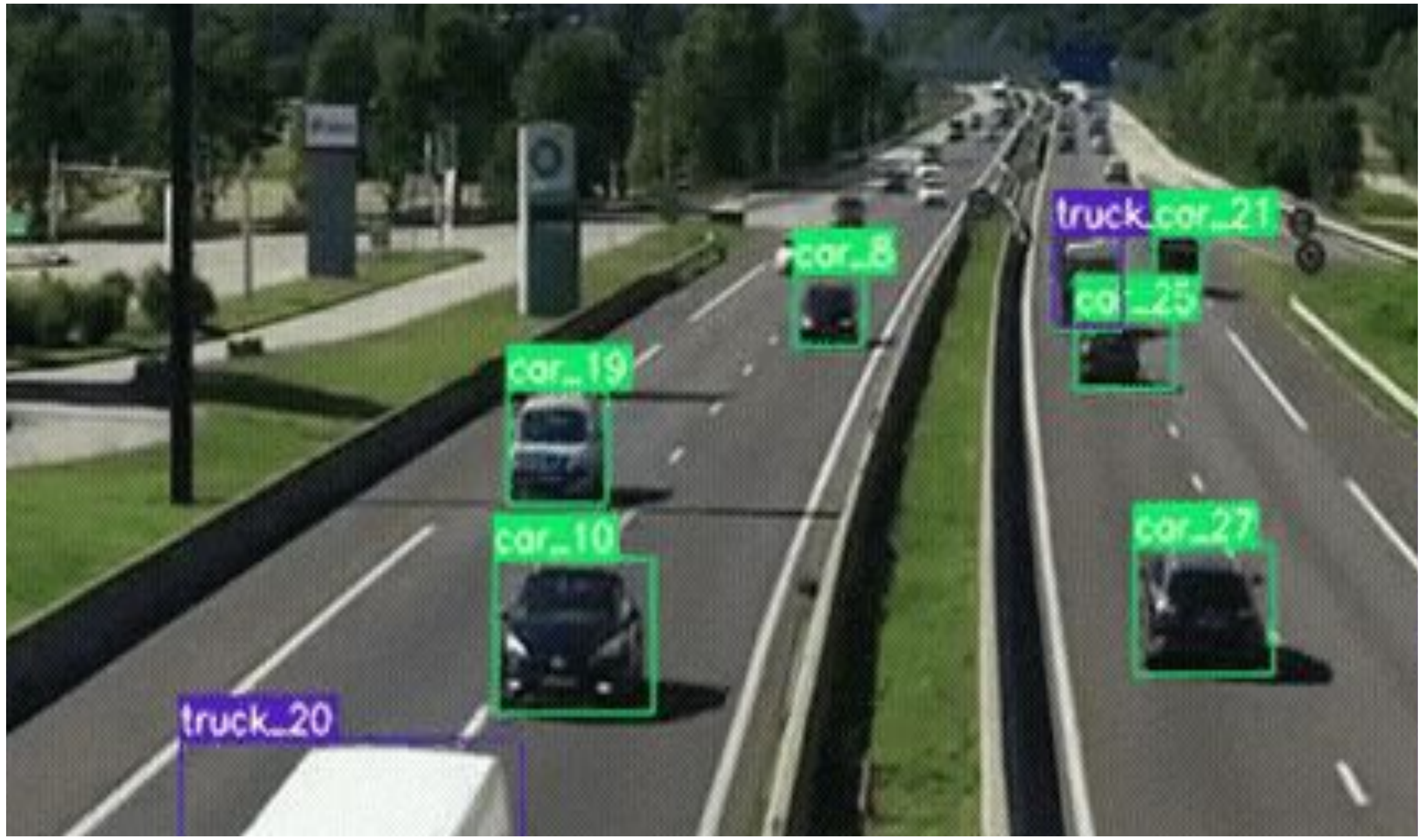
"It's at coordinates (x,y) with this bounding box"

Real AUV Applications:

• Detecting underwater gates 🚪 • Finding specific objects (torpedoes, markers) • Obstacle avoidance • Marine life identification

It's like giving eyes and brain to our robots!

Some Examples!!



Traditional Methods (For the ML Nerds)

Haar Cascades

- ✅ Simple, fast
- ❌ Limited accuracy
- ❌ Only specific objects

HOG + SVM

- ✅ Better accuracy
- ❌ Slow processing
- ❌ Manual feature engineering

R-CNN Family

- ✅ High accuracy
- ❌ Very slow (2-stage)
- ❌ Complex pipeline

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❌ Complex pipeline

YOLO (Look Only Once!!!???)

1. “You Only Look Once”

   YOLO got its name because it detects objects in just one pass through the image—unlike older methods that scan multiple times.

1. Blazing Fast!!

   Processes images in real time (eg. 30FPS - 60FPS), which is great for robots, AUVs, drones and stuff like self driving cars!

1. Looks Once and Detects Many!

   Can detect and classify objects in a given frame, in a single pass! It balances it’s accuracy and speed while the older methods like R-CNN are slower!

YOLO = "You Only Look Once"

The Speed Demon of Object Detection! ⚡

⚡ Super Fast

Real-time detection (30+ FPS)

🤖 Jetson Nano Friendly

Optimized for edge devices

👁 Single Pass

Looks at image once, detects everything

🛠 Practical

Easy to implement and customize

Why and How we use YOLO…

1. Hardware Flexibility

Runs both on our camera (OAK-D), yea our camera has its own processing chip!, and our primary processing unit, The NVIDIA Jetson Nano.

1. Competition-Ready Datasets

Most underwater competitions (e.g., RoboSub, SAUVC) provide labeled datasets (buoys, gates, etc.) — YOLO's Roboflow compatibility makes training a breeze.

1. Future Proof Updates

From YOLOv5 → v8 (or v9 soon!), you get new features (e.g., instance segmentation) *without* rewriting your entire codebase.

The Magic Pipeline:

1. **OAK-D captures** stereo images
2. **Jetson Nano processes** the feed
3. **YOLO model analyzes** each frame
4. **Results displayed** with bounding boxes
5. **Depth info added** from stereo vision

QUICK QUIZ!!!

What makes YOLO special for real-time applications?

A) It's very accurate

B) It processes images very fast

C) It can detect multiple objects in one pass

D) Both B and C

Think about our AUV requirements!

QUICK QUIZ!!!

What makes YOLO special for real-time applications?

A) It's very accurate

B) It processes images very fast

C) It can detect multiple objects in one pass

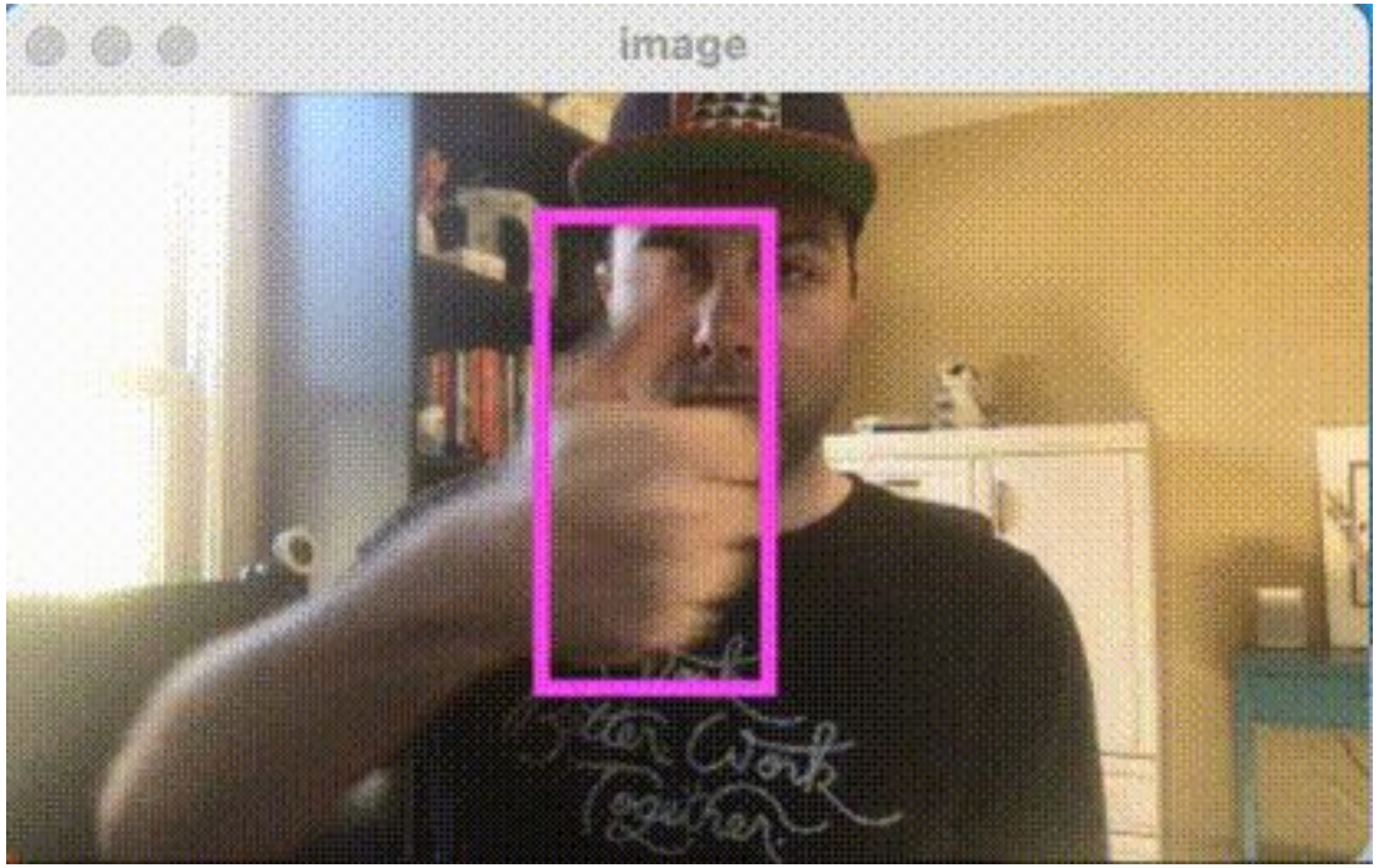
D) Both B and C

Think about our AUV requirements!

Answer: D) Both B and C ✓
YOLO is special because it processes images very fast AND can detect multiple objects in one pass!

How can you train a YOLO model?

Roboflow => Your Dataset Superpower



What is Roboflow?

The easiest way to prepare your training data!

1. Upload Images → 2. Annotate → 3. Label → 4. Export → 5. Train

🌐 Web-based

No software installation needed

🔄 Auto-formats

Generates different export formats

📈 Data Augmentation

Built-in image enhancement

👥 Team Collaboration

Work together on datasets

How can you train a YOLO model?

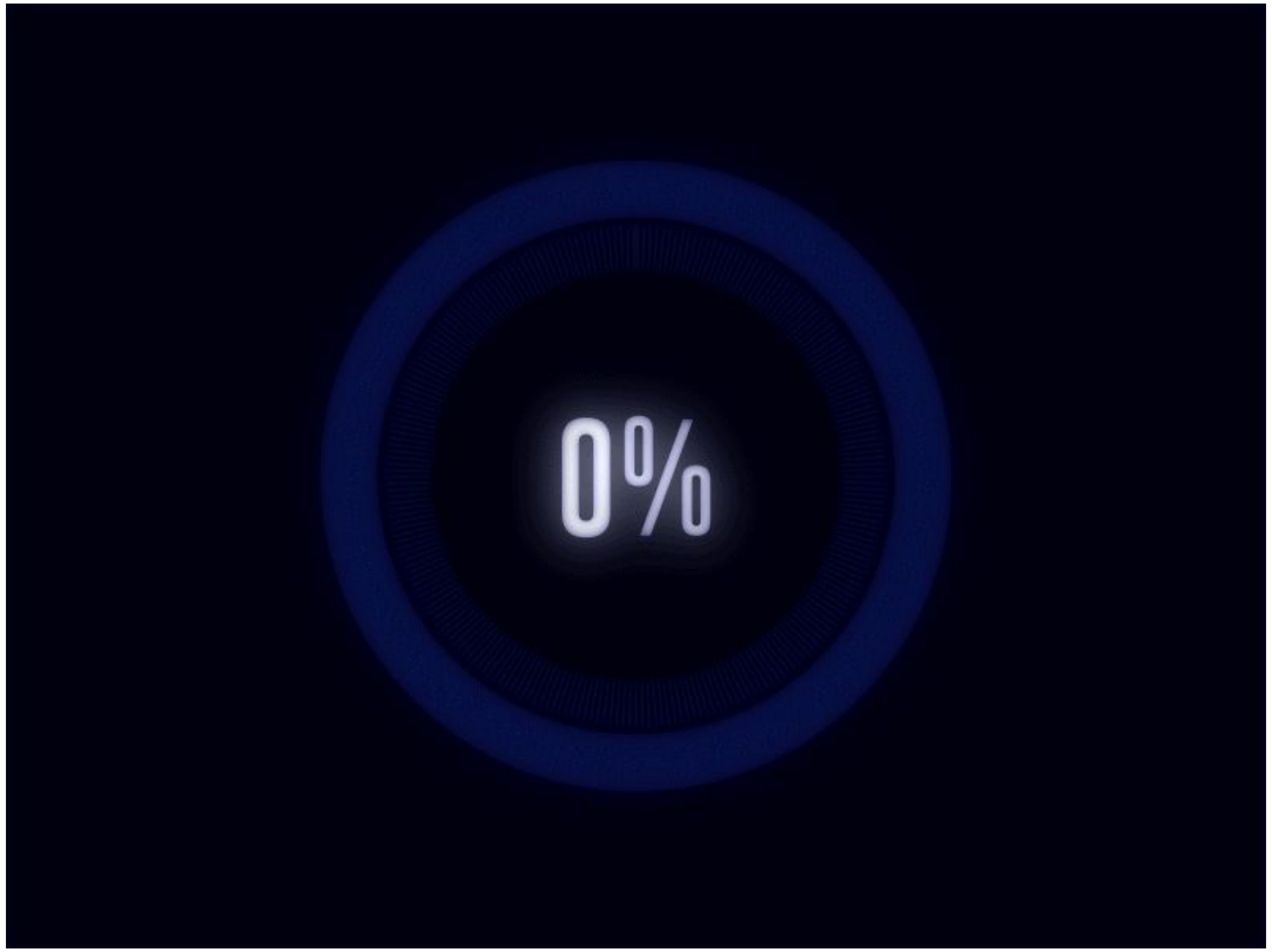
STEP - 1

Dataset Prep & Labeling in Roboflow

- Collect Images: Gather a dataset (e.g., 100+ images of your target object).
- Upload to Roboflow:
  - Create a project → Upload images → Use Roboflow's annotation tool to draw bounding boxes around objects.
- Preprocess & Augment:
  - Apply auto-orient, resize (640x640), and augmentations (flips, blur) to improve model robustness.

How can you train a YOLO model?

STEP - 2



Export & Train with YOLO v5/YOLO v8

- Export Dataset: In Roboflow, go to "Generate" → Select YOLO v5 PyTorch format → Download the dataset with train/val/test splits.
- Use YOLO v5 Colab Notebook:
  - Open the official YOLO v5 Training Notebook
  - Upload your Roboflow dataset → Paste the dataset.yaml path from Roboflow
- Start Training:
  - Run “**!python train.py --data /path/to/dataset.yaml --weights yolov5s.pt --epochs 100**”
  - Monitor training progress and download the trained best.pt model file

How can you train a YOLO model?

STEP - 3

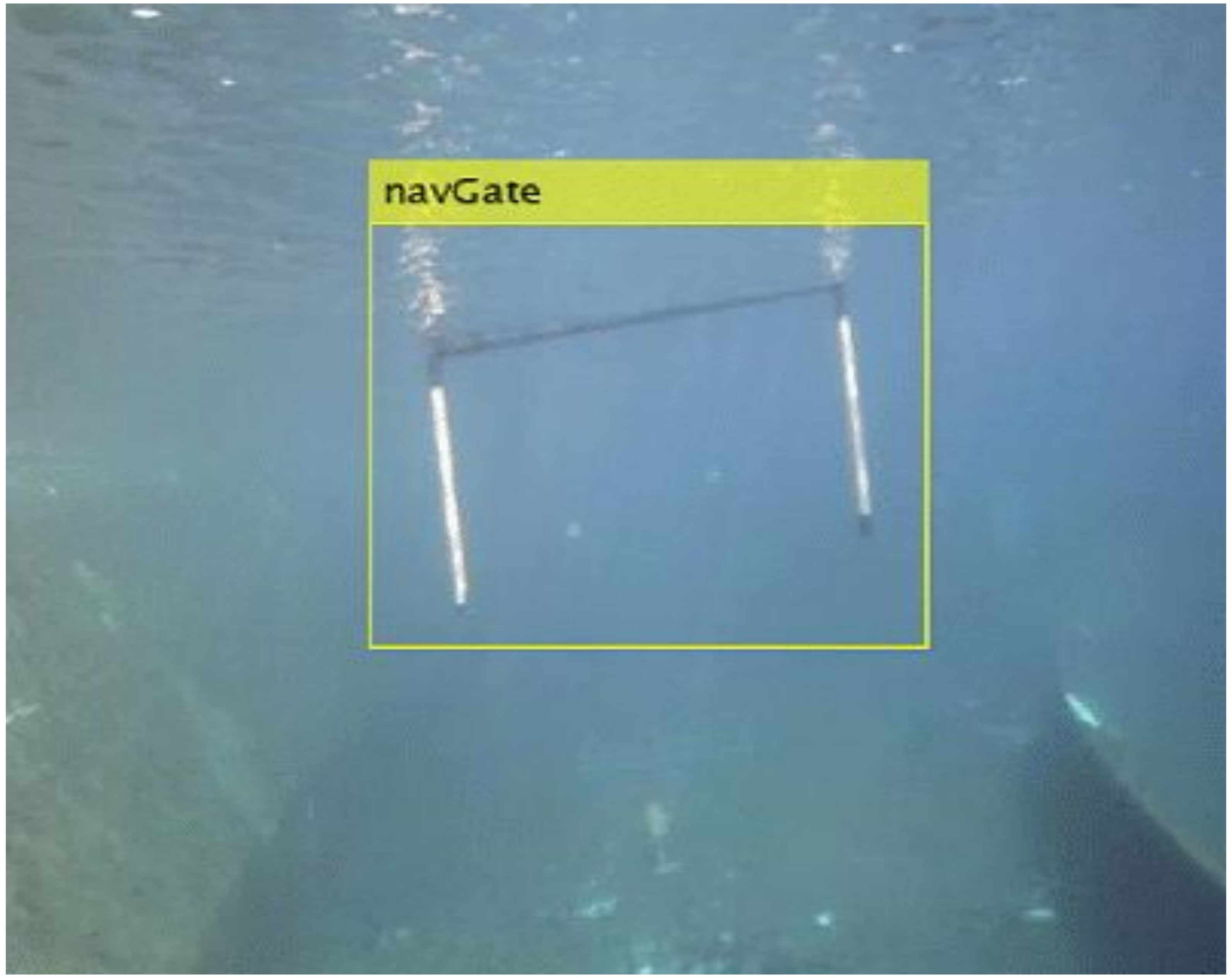
Download Trained Mode and Set Up Detection

Script:

- Clone YOLO v5 repo locally → Install requirements → Place your best.pt in the repo
- Use: python detect.py --weights best.pt --source 0 (0 = webcam)

Real-time Testing:

- Run the script → Point webcam at your trained object → See live bounding boxes and confidence scores
- Fine-tune confidence threshold (--conf 0.5) for better detection accuracy




Live Demo

Roadmap?? Just keep exploring bro!

Some decent resources!

- https://www.youtube.com/watch?v=r0RspiLG260
- https://www.geeksforgeeks.org/yolo-you-only-look-once-real-time-object-detection/
- https://www.youtube.com/watch?v=fu2tfOV9vbY

Step 1 Master Python fundamentals and OpenCV basics

Step 2 Understand YOLO concepts and run pre-trained models

Step 3 Set up development environment (Python, PyTorch, YOLOv5/v8)

Step 4 Create your first custom dataset using Roboflow

Step 5 Train your first custom YOLO model on your dataset

Step 6 Deploy and test your model with webcam/real-time applications

You got the idea! Now build and explore!

"Learning by doing is the best way to learn anything new"

Project 1: Basic Object Counter

Count people, cars, or objects in images using pre-trained YOLO

Project 2: Real-time Webcam Detector

Use your webcam to detect common objects (person, phone, bottle) live

Project 3: Custom Single-Class Detector

Train YOLO to detect one specific object (your face, pet, or favorite item)

Project 4: Multi-Class Custom Dataset

Create dataset with 3-5 different objects and train a multi-class detector

Project 5: Smart Security System

Build an alert system that detects intruders and sends notifications

Project 6: Advanced Application

Sports analysis, traffic monitoring, or industrial quality control system

Start with Project 1 and work your way up!

Career Prospects

💻 Computer Vision Engineer
NVIDIA, Intel, Qualcomm

🤖 Robotics Engineer
Boston Dynamics, Tesla

🔬 AI Research
Top universities and labs

🚗 Autonomous Systems
Automotive, aerospace, marine

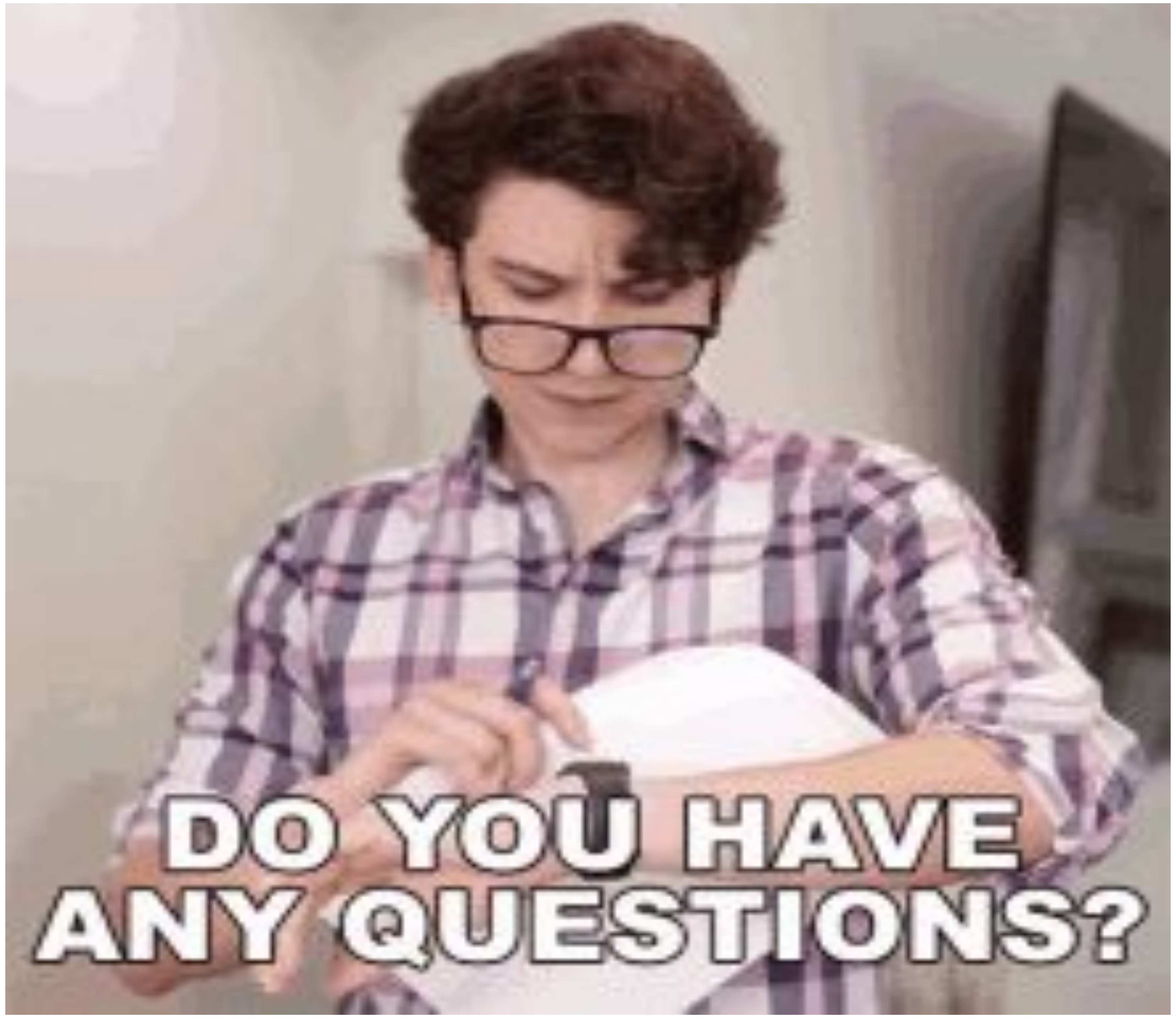
🚀 Startup Opportunities
AI/ML companies

🛡️ Defense & Aerospace
Underwater robotics, surveillance

💰 Salary Range:
Entry Level: ₹8-15 LPA
Mid Level: ₹20-40 LPA
Senior Level: ₹50+ LPA

QnA

Any Doubts??



Connect with us

AUV Society IIITDM Kancheepuram!!!

Thank You

gotta say thank you for their time